深蓝学院
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Embodied Intelligence：Robotic Arm Manipulation
具身智能：机械臂操作

Lecture 3：Visual Grasping

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Course Outline

□ 具身智能：机械臂操作

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Introduction

- 抓取任务介绍
- 硬件配置
- 软件环境配置

Course 2
机械臂运动学

- 正运动学
- 逆运动学
- 路径规划

Course 3
经典抓取

- 目标检测
- 规则抓取

Course 4
强化学习

- 强化学习算法
- 仿真器配置
- 强化学习训练

Course 5
VLA模仿学习

- 遥操作配置
- 网络模型结构
- 模型训练使用

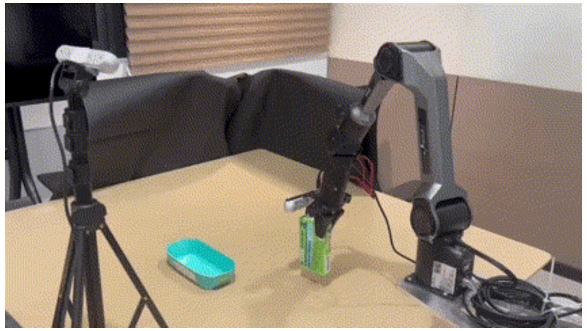
Course 6
前沿分享

- 狗+机械臂联合强化学习
- 3DGS仿真器抓取任务Sim2Real

Day I

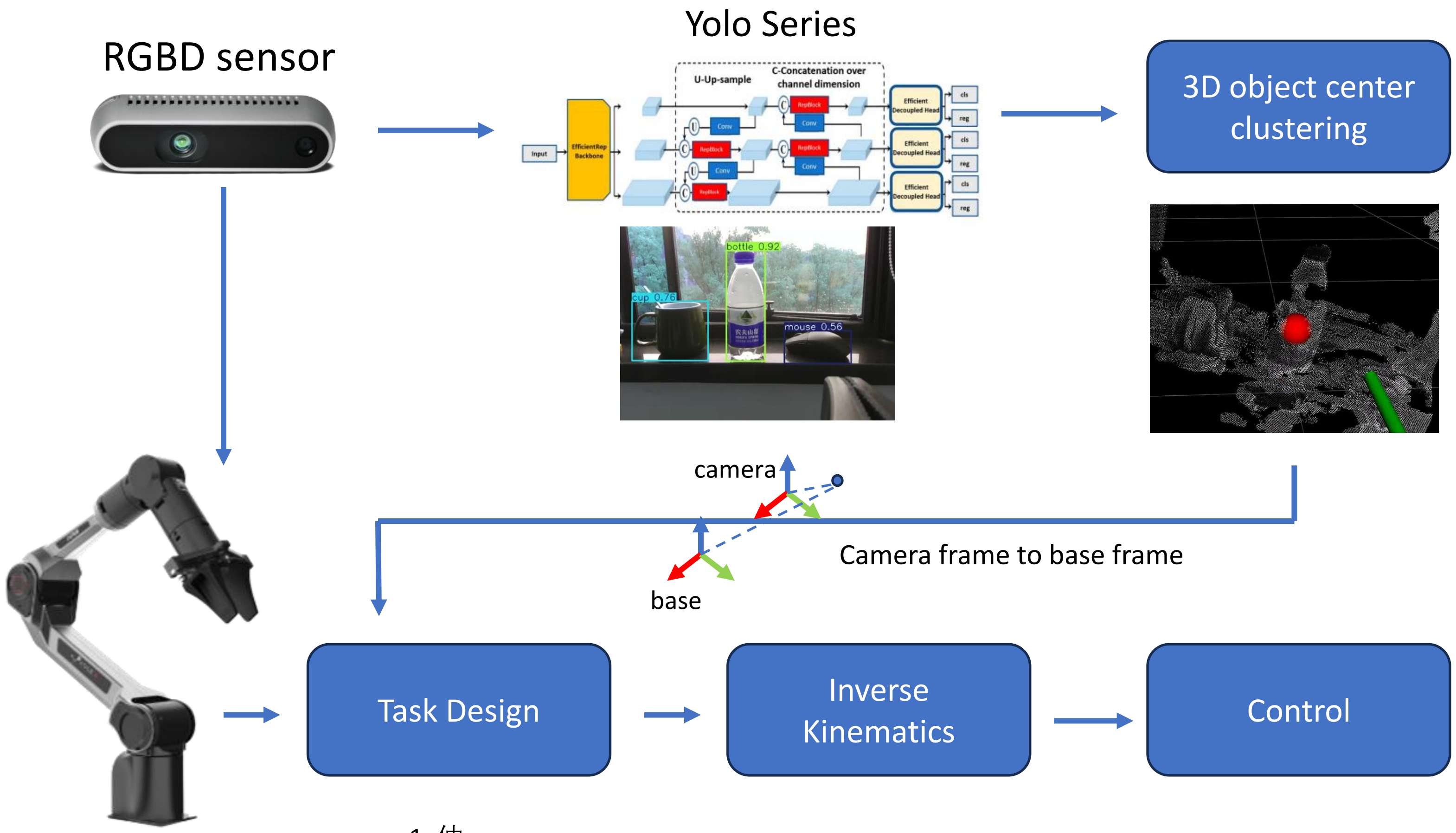
Day II

Day III



Content

Visual Grasping Pipeline

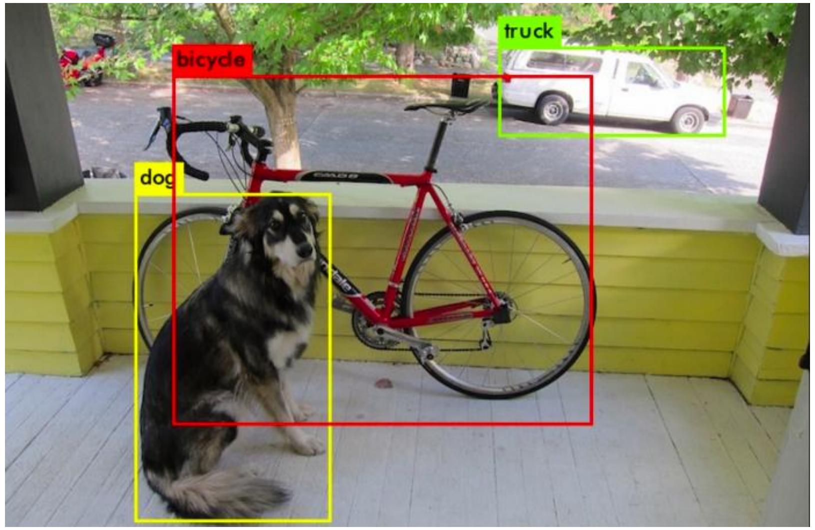


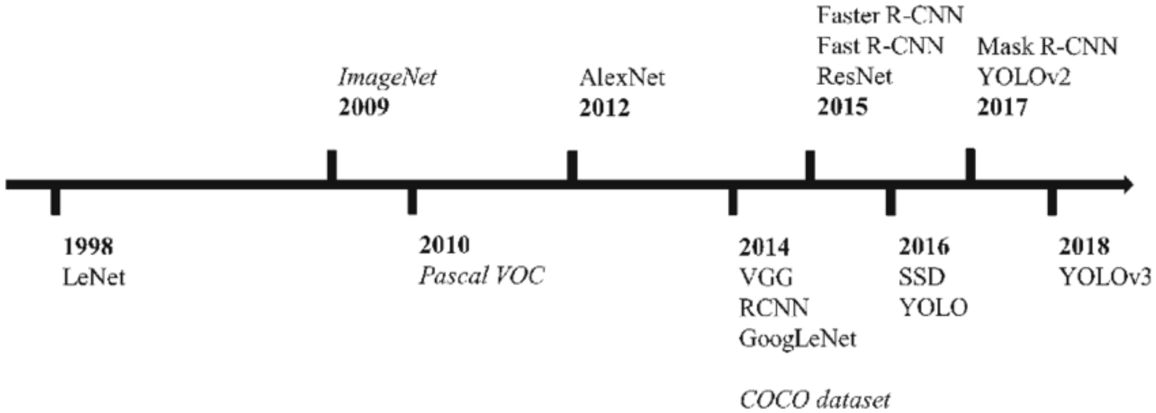
1. 伸
2. 抓
3. 回

Content

1. Visual Grasping Pipeline
2. Visual Detection
3. Visual Geometry (2D -> 3D)
4. Grasping Task Design
5. Assignment

Visual Detection





YOLO 版本	架构创新	训练策略	优化技术
YOLOv1	简化的 CNN 骨干网络，基本的边界框预测	几何变换，色调抖动	随机梯度下降，非极大值抑制 (NMS)
YOLOv2	DarkNet-19 骨干网络，K-means 聚类用于锚框优化	微调，预训练权重	带动量的 SGD，超参数调优，Adam 优化器
YOLOv3	DarkNet-53，多尺度检测，残差连接	Mix-up，数据增强，噪声	非极大值抑制，阈值处理，多尺度目标检测
YOLOv4	CSPDarkNet-53，PANet，马赛克数据增强	迁移学习，知识蒸馏	广义 IoU，焦点损失，动态量化
YOLOv5	CSPNet，动态锚框优化，轻量级架构	马赛克，CutMix，早停	训练后量化，滤波器剪枝，低秩近似
YOLOv6	PANet，CSPDarkNet53	对抗训练，领域特定的数据增强	IoU 损失，置信度阈值处理，多尺度融合
YOLOv7	EfficientRep 骨干网络，动态标签分配	微调，对抗补丁检测	神经架构搜索 (NAS)，量化，梯度裁剪
YOLOv8	路径聚合网络，动态核注意力	对抗训练，数据增强	动量和 Adam 优化器，训练后量化
YOLOv9	多级辅助特征提取	微调，领域特定的数据增强	GELAN 模块，资源受限系统的深度监督
YOLOv10	轻量级分类头和分离的空间与通道变换	无 NMS 训练，双重标签分配	在下采样阶段通过分离的空间与通道变换提高整体效率
YOLOv11	在骨干网络中引入 C3k2 块，并使用 C2PSA 增强空间注意力	微调，Mix-up，数据增强，自适应梯度裁剪	量化，随机梯度下降 (SGD)
YOLOv12	主干网络引入区域注意力模块（area attention, A2），并引入了残差高效层聚合网络（R-ELAN），	Mosaic、Mixup 和复制粘贴增强等	FlashAttention，卷积实现注意力)
YOLO-World	CLIP-enhanced检测头，可重参数化VL-PAN	区域文本对比训练，动态提示学习	开放词汇蒸馏，跨模态注意力

Visual Detection

- YOLO v1

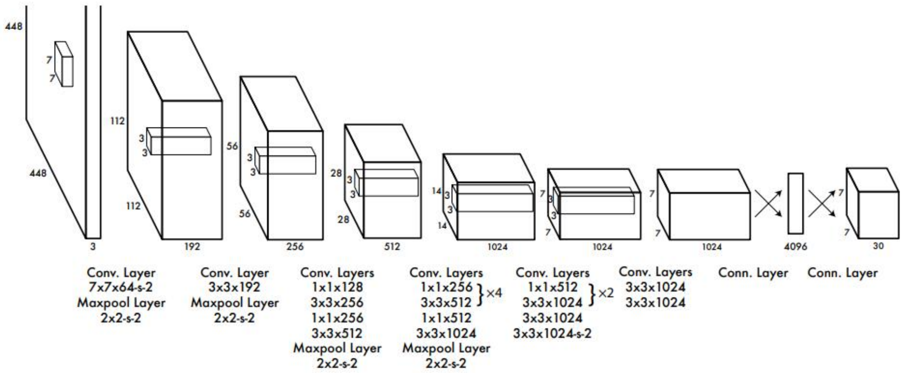
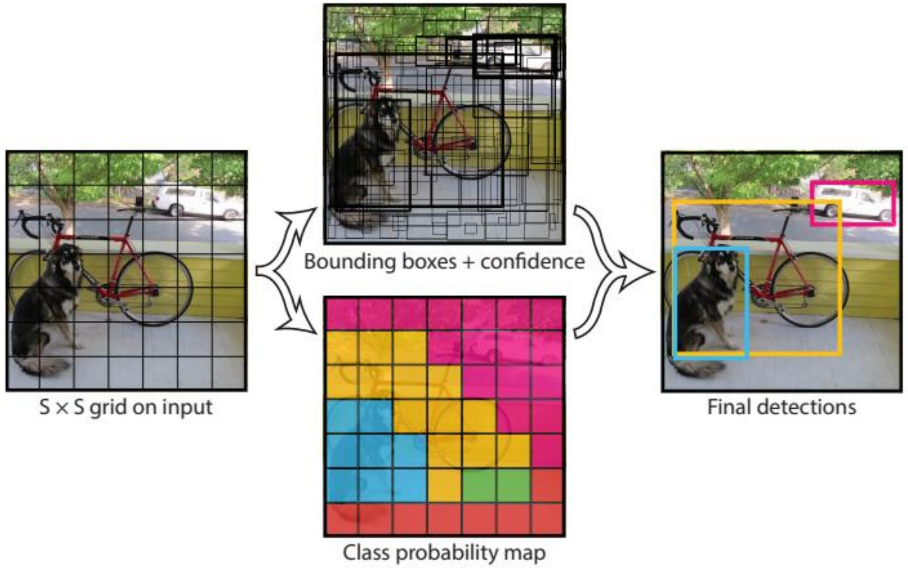
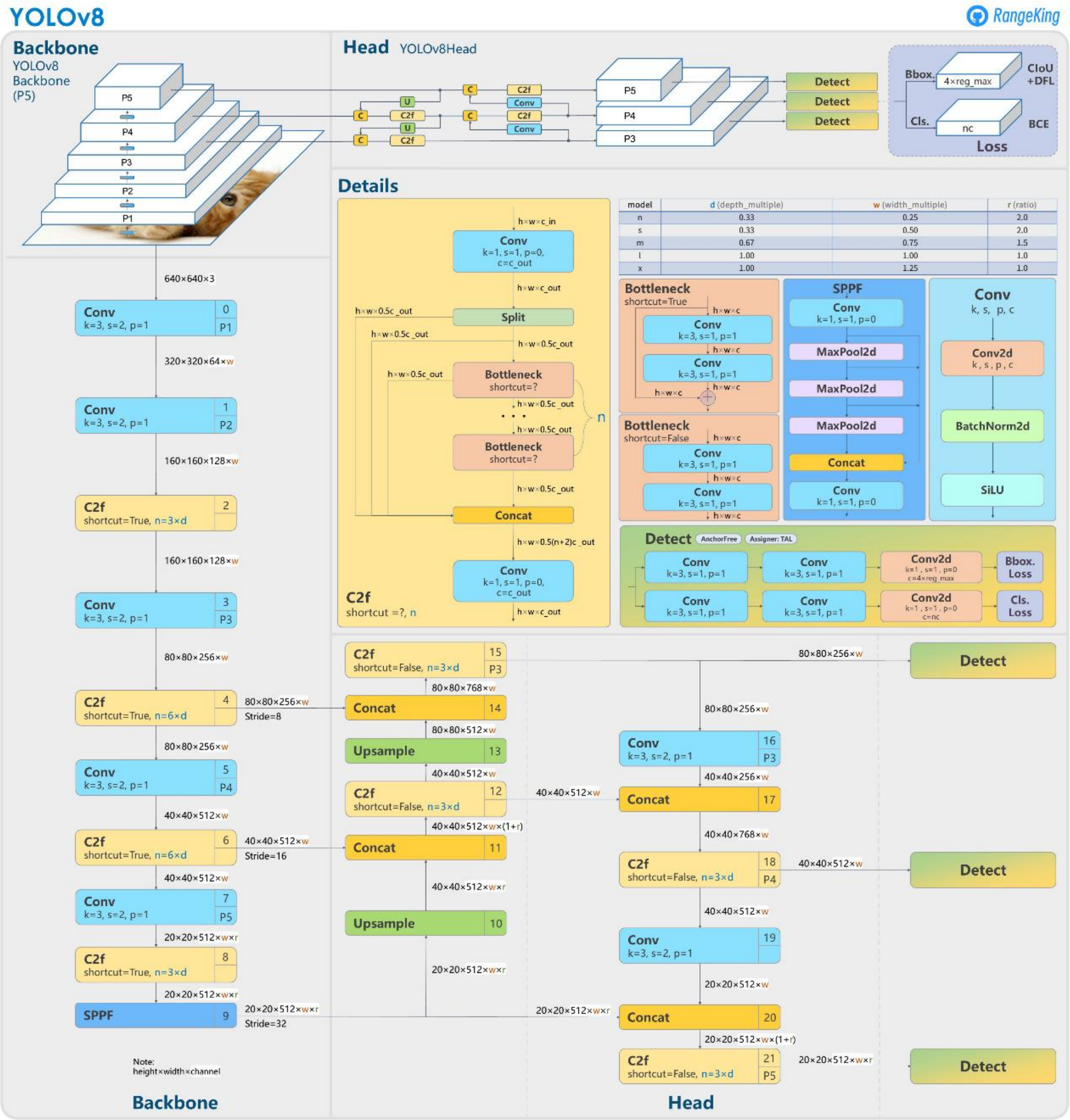


Figure 3: The Architecture. Our detection network has 24 convolutional layers followed by 2 fully connected layers. Alternating 1×1 convolutional layers reduce the features space from preceding layers. We pretrain the convolutional layers on the ImageNet classification task at half the resolution (224×224 input image) and then double the resolution for detection.



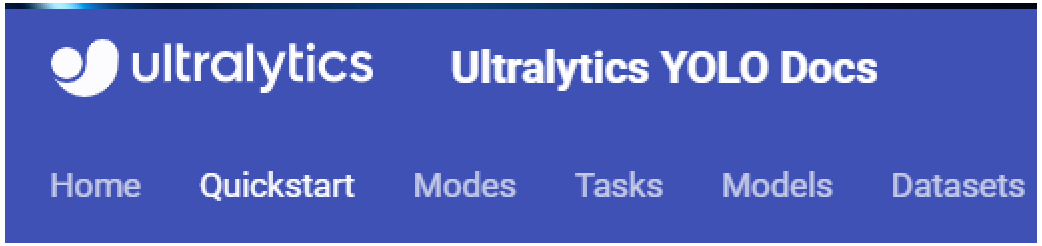
Visual Detection

- YOLO v8



Visual Detection

- YOLO Deployment and Installation
  - https://docs.ultralytics.com/quickstart/



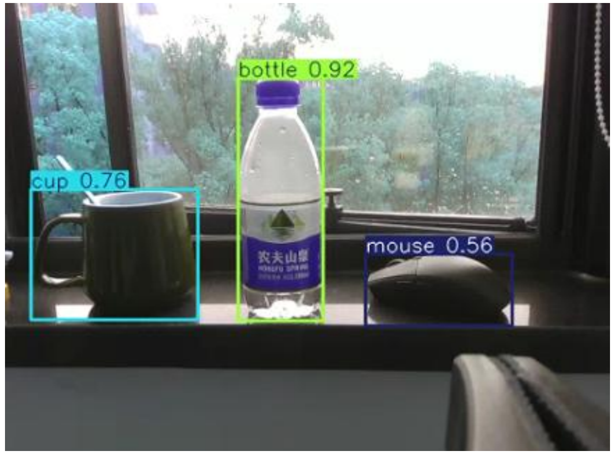

```
# Install the ultralytics package from PyPI
pip install ultralytics
```

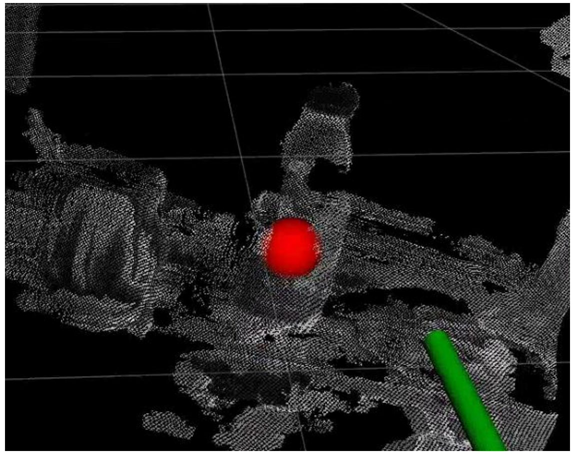
```python
from ultralytics import YOLO

# Load a model
model = YOLO("yolo11n.pt")  # pretrained YOLO11n model

# Run batched inference on a list of images
results = model(["image1.jpg", "image2.jpg"])  # return a list of Results objects

# Process results list
for result in results:
    boxes = result.boxes  # Boxes object for bounding box outputs
    masks = result.masks  # Masks object for segmentation masks outputs
    keypoints = result.keypoints  # Keypoints object for pose outputs
    probs = result.probs  # Probs object for classification outputs
    obb = result.obb  # Oriented boxes object for OBB outputs
    result.show()  # display to screen
    result.save(filename="result.jpg")  # save to disk
```

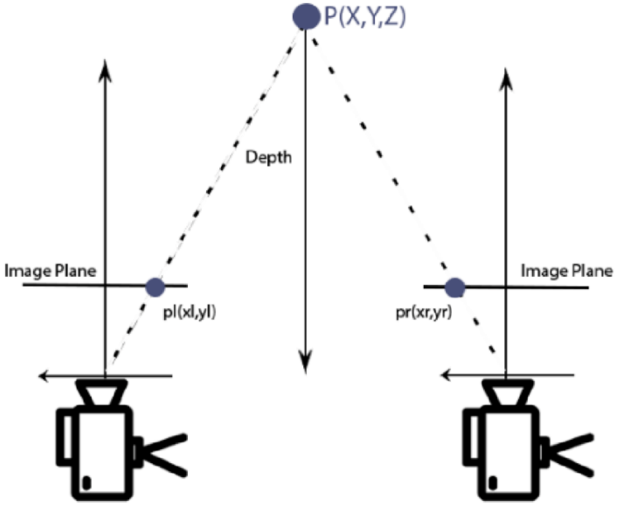
Content

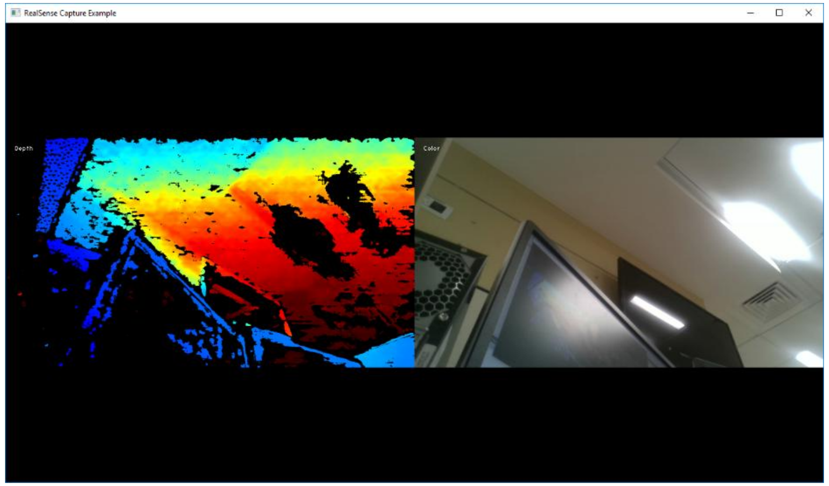
Visual Geometry (2D -> 3D)

- Realsense camera
    - Depth estimation



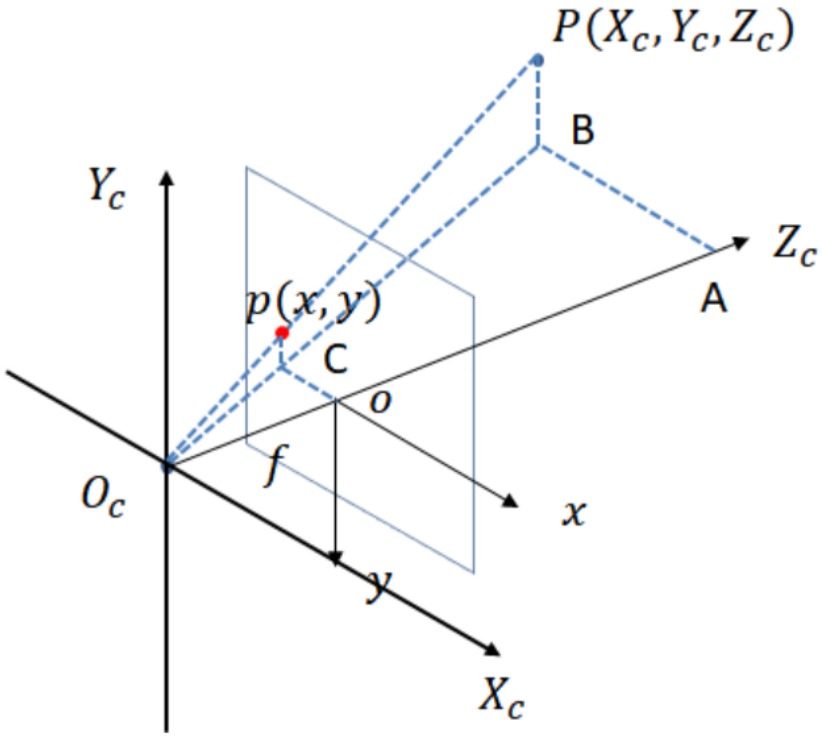






Visual Geometry (2D -> 3D)

- Camera Projection



$\Delta ABO_c \sim \Delta oCO_c$

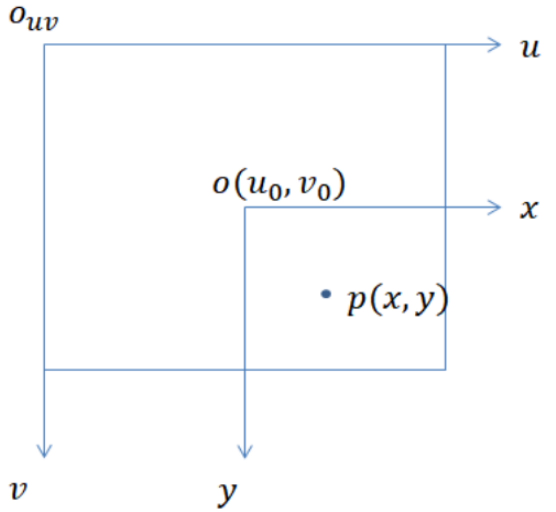
$\Delta PBO_c \sim \Delta pCO_c$

$$\frac{AB}{oC} = \frac{AO_c}{oO_c} = \frac{PB}{pC} = \frac{X_c}{x} = \frac{Z_c}{f} = \frac{Y_c}{y}$$

$$x = f\frac{X_c}{Z_c}, y = f\frac{Y_c}{Z_c}$$

$$Z_c\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} X_c \\ Y_c \\ Z_c \\ 1 \end{bmatrix}$$

图像坐标系->像素坐标系



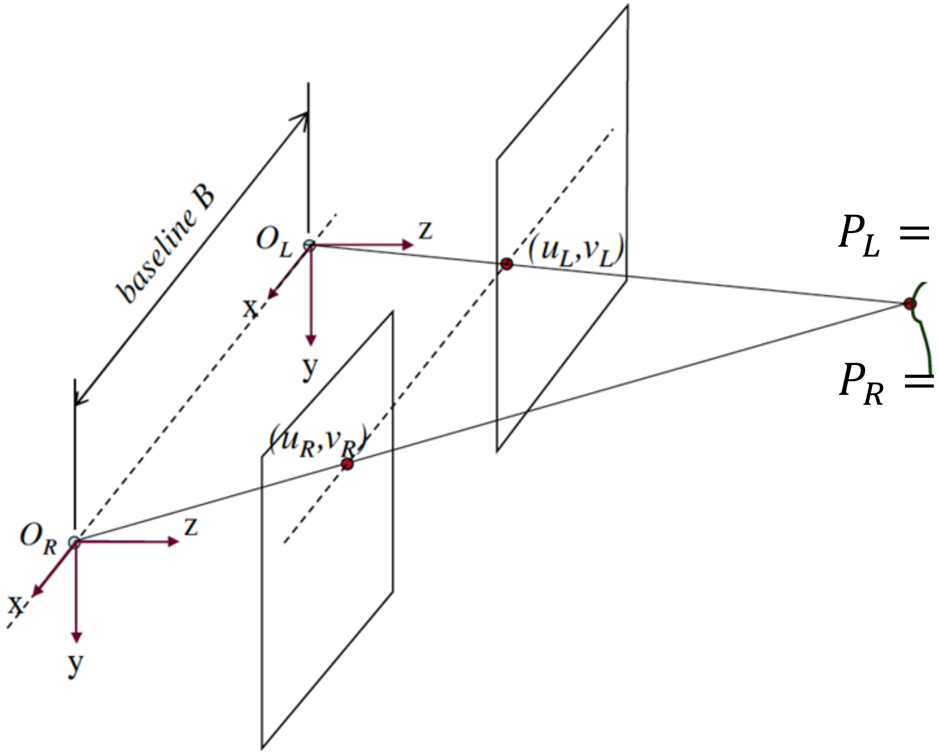
$$u = \frac{fX_c}{Z_c} + u_0$$

$$v = \frac{fY_c}{Z_c} + v_0$$

$$Z_c\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} f & 0 & u_o \\ 0 & f & v_o \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} X_c \\ Y_c \\ Z_c \end{pmatrix}$$

Visual Geometry (2D -> 3D)

- Stereo Depth Estimation



$P_L = (X, Y, Z)$

$P_R = (X - B, Y, Z)$

$$(x_L, y_L) = \left(f\frac{X}{Z}, f\frac{Y}{Z}\right)$$

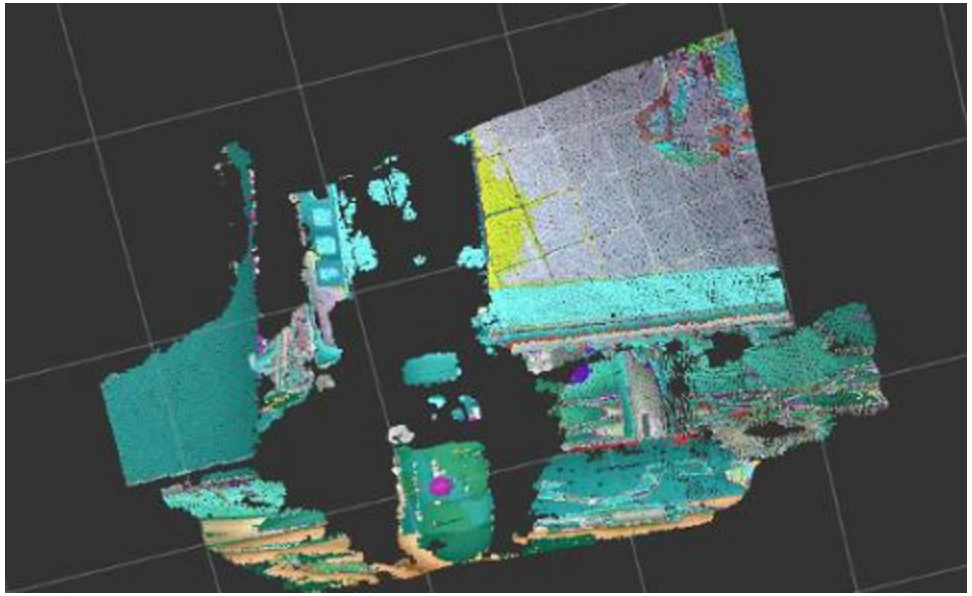
$$(x_R, y_R) = \left(f\frac{X - B}{Z}, f\frac{Y}{Z}\right)$$

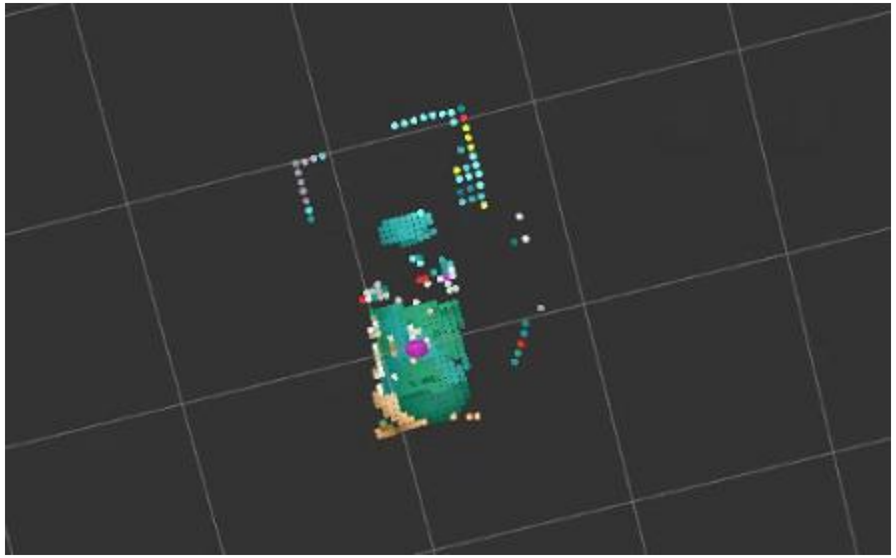
Disparity:

$$d = u_L - u_R = x_L - x_R = f\frac{B}{Z} \qquad Z = f\frac{B}{d}$$

Visual Geometry (2D -> 3D)

- Realsense camera
  - Depth estimation
  - Project (uv -> xyz)
  - Point cloud clustering

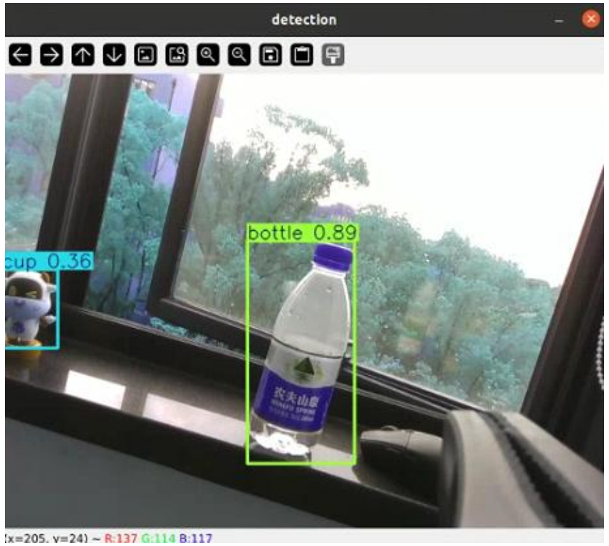


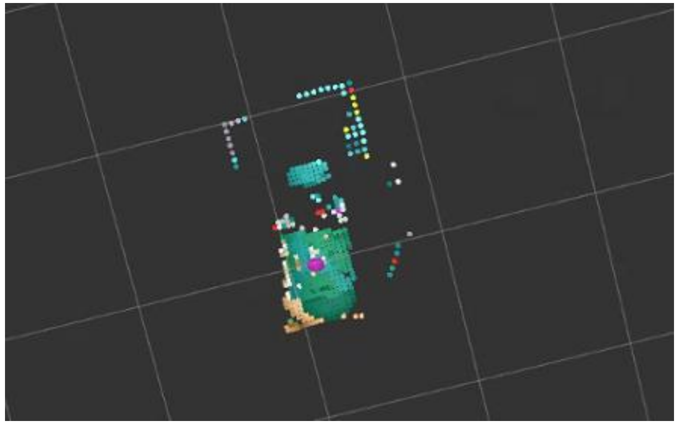


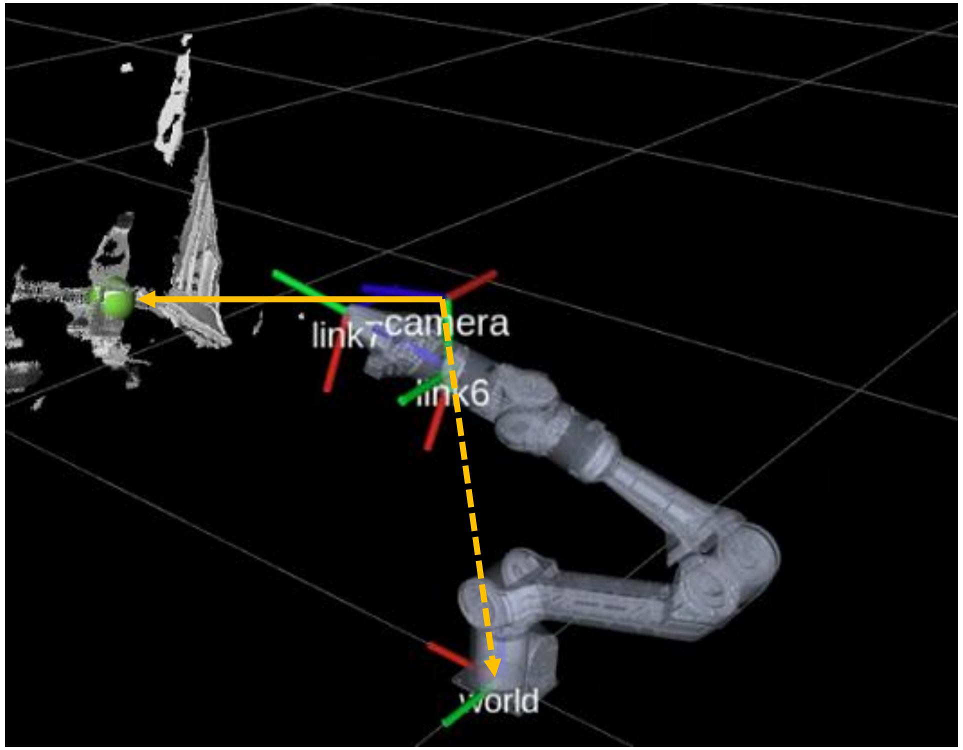
Content

Grasping Task Design

- Coordinate Transformation
  - Camera frame point ${}^{c}p$ -> base(world) frame point ${}^{w}p$
  - Inverse Kinematics: $\theta_1, \theta_2 \ldots \theta_n = f^{-1}({}^{w}p, {}^{w}R)$
  - State machine: 开抓-> 伸手->握爪-> 回来







Content

1. Visual Grasping Pipeline
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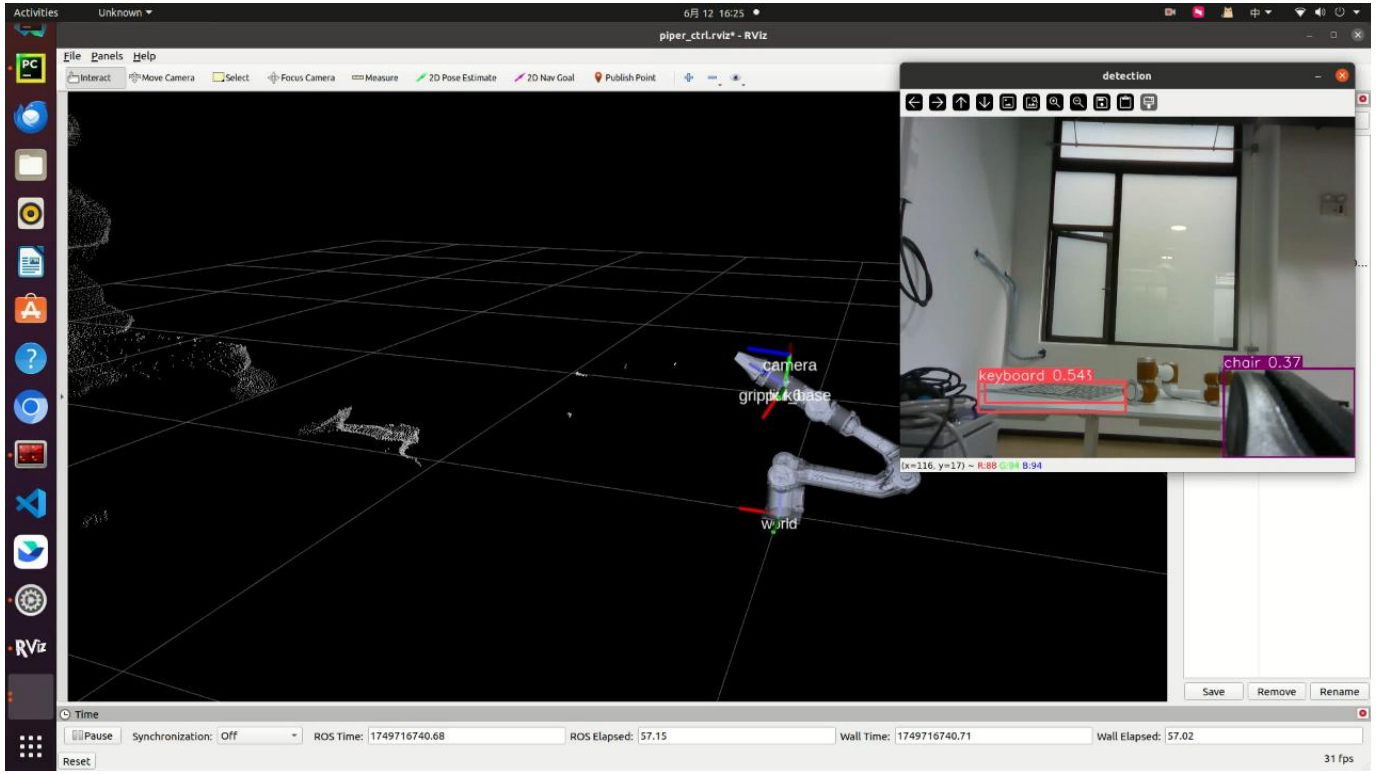
4. Grasping Task Design
5. Project 2

Project 2: Overview

机械臂抓取



RVIZ界面

Project 2

- Yolo setup

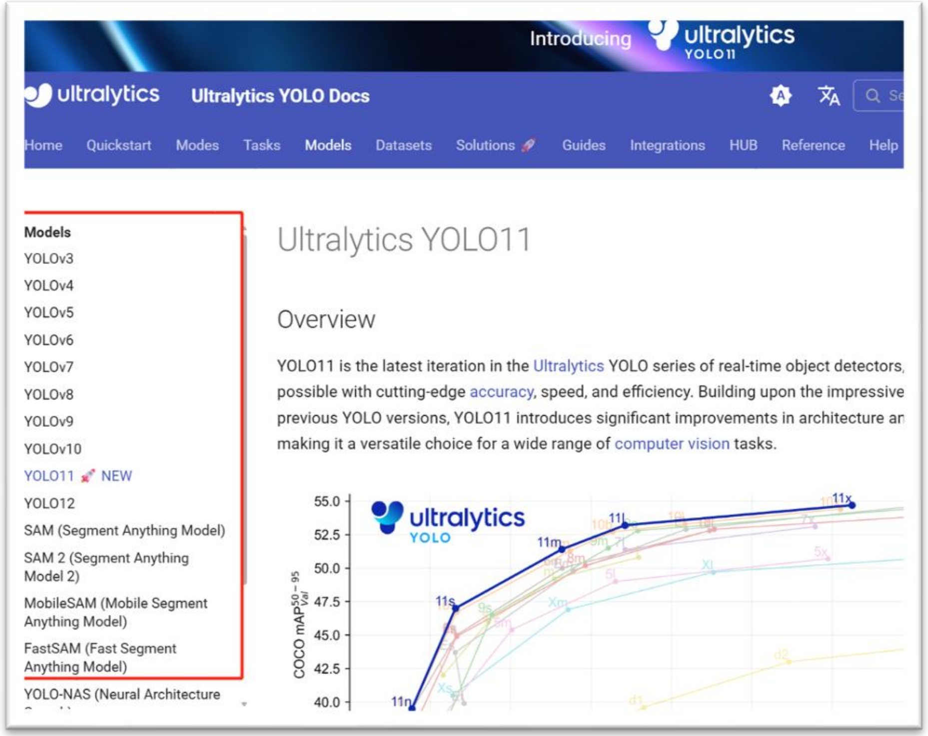
    - https://docs.ultralytics.com/models/yolo11/
        - pip install ultralytics
        - python test_yolo.py

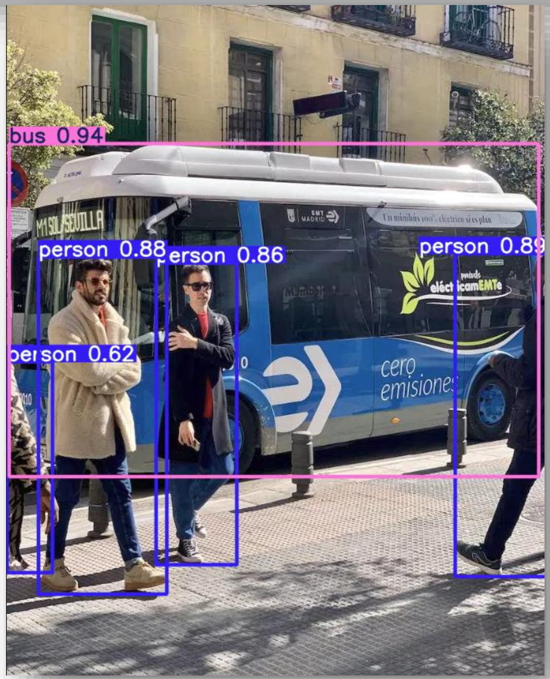
```python
from ultralytics import YOLO

# Load a pretrained YOLO model
model = YOLO("yolo11n.pt")

# Perform object detection on an image
results = model("bus.jpg")

# Visualize the results
for result in results:
    result.show()
```





Project 2

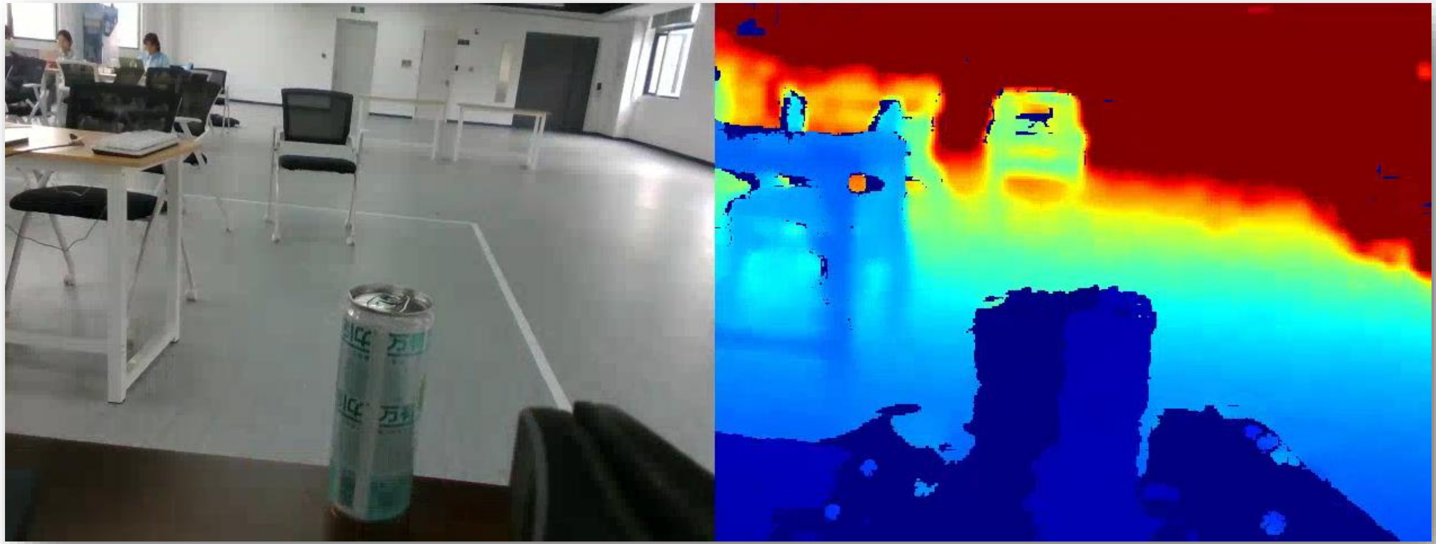
- Real sense setup

  - Install lib

    ```
    pip install pyrealsense2
    ```

  - Test

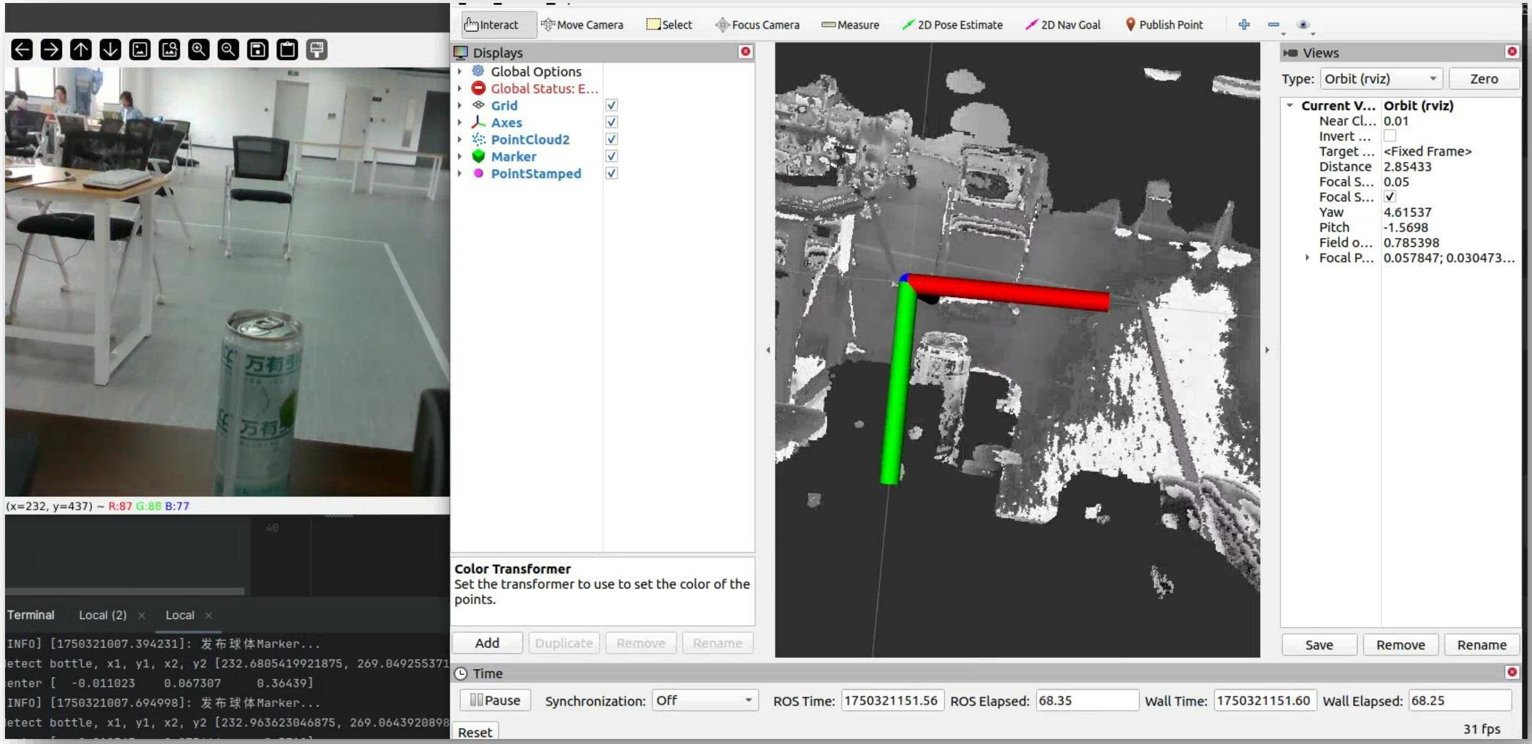
    ```
    python test_realsense.py
    ```



Project 2

➢ 2D Depthmap -> 3D Pointcloud

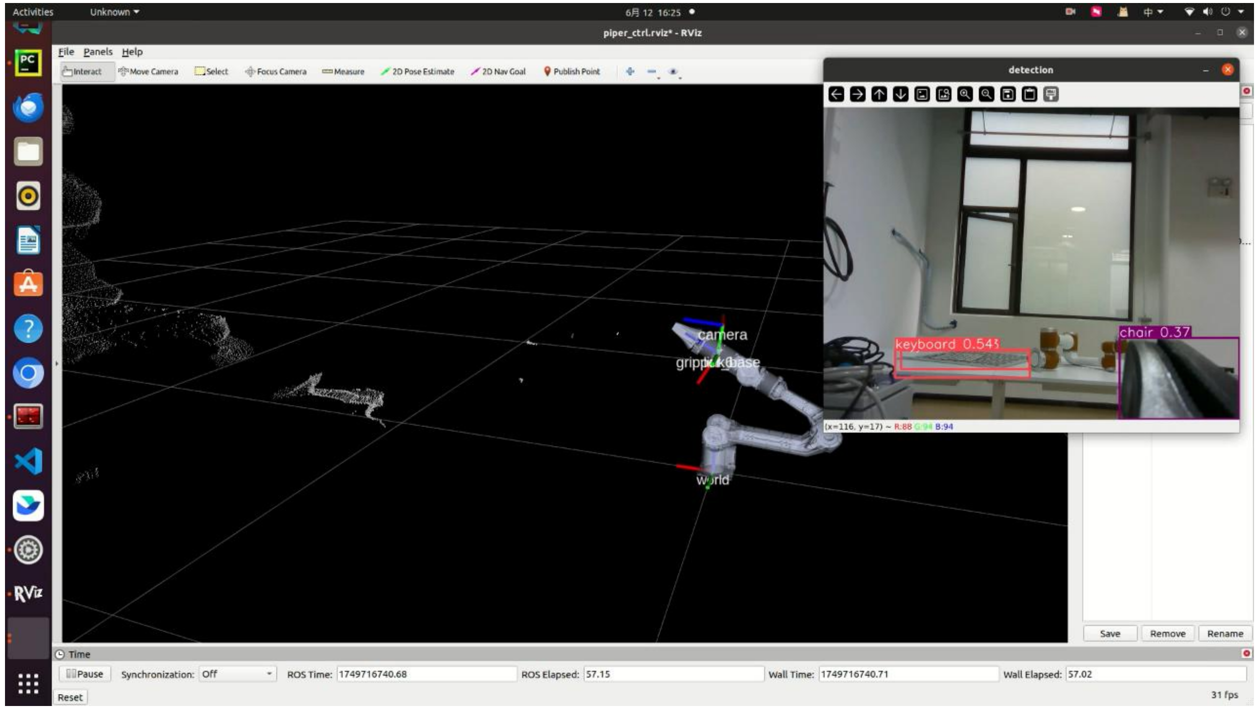
- python test_depth_2_pointcloud.py



Project 2

- ROI point extraction
  - python realsense_yolo_roi.py

- Control arm to reach that point and grasp
  - Python grasp_action.py



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感谢聆听

Thanks for Listening!